Detection and Analysis of GPS L1-Band Radio Frequency Interference Using Spaceborne Global Navigation Satellite System Reflectometry Receivers

Mohammad M. Al-Khaldi, *Member, IEEE*, Joel T. Johnson, *Fellow, IEEE*, Darren S. McKague, *Member, IEEE*, Scott Gleason, Frederick S. Policelli, Rajat Bindlish, *Senior Member, IEEE*, Dorina Twigg, and Anthony Russel

***Abstract*— We report a study of radio frequency interference (RFI) in the GPS L1-band (1575.42 ± 2 MHz) using spaceborne global navigation satellite system reflectometry (GNSS-R) data. An examination of the "noise" levels reported by both Spire, Inc. and Cyclone GNSS (CYGNSS) standard Level 1 products is first presented, which shows clear evidence of RFI contributions. Data from 498 acquisitions in CYGNSS's special raw intermediate frequency (I/F) mode are then used to examine RFI source properties in greater detail. Both kurtosis and cross-frequency algorithms are applied with the raw I/F data to detect the presence of RFI. The results suggest that up to 25% of the acquisitions considered contain RFI, with both "narrow" and "wideband" sources of varying amplitudes observed. The results provide insights into RFI source properties in the GPS L1-band as well as methods by which the RFI can be detected.**

***Index Terms*— Bistatic radar systems, CubeSats, global navigation satellite system reflectometry (GNSS-R), radio frequency interference (RFI), rough surface scattering, signals of opportunity systems, SmallSats.**

I. Introduction

WHILE the primary purpose of global navigation satellite systems (GNSSs) is navigation and positioning, GNSS measurements also provide vital Earth monitoring capabilities such as surface deformation mapping [1], [2], integrated water vapor monitoring [3], and seismology [4]. GNSS reflectometry (GNSS-R) has further been applied for sensing ocean wind speeds [10], [11], surface soil moisture [12], [13], [14], [15], vegetation biomass [16], and inundation dynamics [17], [18], [19] through the measurements of U.K.-Disaster Monitoring Constellation (U.K.-DMC) satellite [5], TDS-1 (Technology Demonstration Mission) [6], Cyclone GNSS (CYGNSS) [7], and FengYun-3E [8] missions as well as the imminent launch of ESA's twin HydroGNSS satellite receivers [9].

While GNSS measurements are performed in multiple frequency bands, the Federal Communications Commission (FCC) and International Telecommunication Union (ITU) [20] designated L1-band that extends from 1559 to 1610 MHz (a portion of the 1–2-GHz L-band range) has been widely used for both navigation and sensing applications, particularly within the 1575.42 ± 2 MHz range transmitted by the global positioning system constellation [30]. Recent studies have indicated a presence of radio frequency interference (RFI) within the L1-band [31], [32], [33], [34], [35], [36], [37], [38]. RFI contributions are undesirable due to their potential to degrade navigation and/or sensing performance. While the detection, mitigation, and geolocation of terrestrial L-band RFI in the 1400–1427-MHz band reserved for radio astronomy and microwave radiometry have been extensively demonstrated in the context of the Aquarius [21], [22], Soil Moisture and Ocean Salinity (SMOS) [23], [24], [28] and Soil Moisture Active Passive (SMAP) [26], [27], [28], [29] missions, the detection or characterization of RFI in the GPS L1-band using spaceborne observations has received less attention, particularly in terms of examining detailed RFI source properties.

This work examines the use of spaceborne GNSS-R measurements for the detection of GPS L1 interference originating from terrestrial transmitters. Section II provides a review of the operating principles of GNSS-R systems and an initial examination of the noise environment in which they operate using commercial GNSS-R data distributed by the National Aeronautics and Space Administration (NASA's) Commercial Satellite Data Acquisition Program (CSDA) [39], [40], [41] as well as NASA's 2016 CYGNSS Earth Venture Mission. The use of a special CYGNSS data acquisition mode for RFI detection is then presented in Section II-C. Results from the application of both kurtosis and cross-frequency RFI detection algorithms are then discussed in Sections III and IV, respectively. A summary of observed trends, lessons learned, and other final conclusions is then provided in Section V. While questions relating to the geolocation of transmitters and

Received 1 April 2025; revised 28 May 2025 and 21 June 2025; accepted 26 June 2025. Date of publication 30 June 2025; date of current version 9 July 2025. This work was supported in part by National Aeronautics and Space Administration (NASA) Science Mission Directorate through the University of Michigan under Contract NNL13AQ00C, in part by NASA Commercial Satellite Data Acquisition (CSDA) Program under Grant 80NSSC23K1117 and Grant 80NSSC23K0320, and in part by the Allocation of Computing Resources from Ohio Supercomputer Center. *(Corresponding author: Mohammad M. Al-Khaldi.)*

Mohammad M. Al-Khaldi and Joel T. Johnson are with the Department of Electrical and Computer Engineering and ElectroScience Laboratory, The Ohio State University, Columbus, OH 43210 USA (e-mail: al-khaldi.2@osu.edu).

Darren S. McKague is with the Department of Climate and Space Sciences and Engineering, University of Michigan, Ann Arbor, MI 48109 USA.

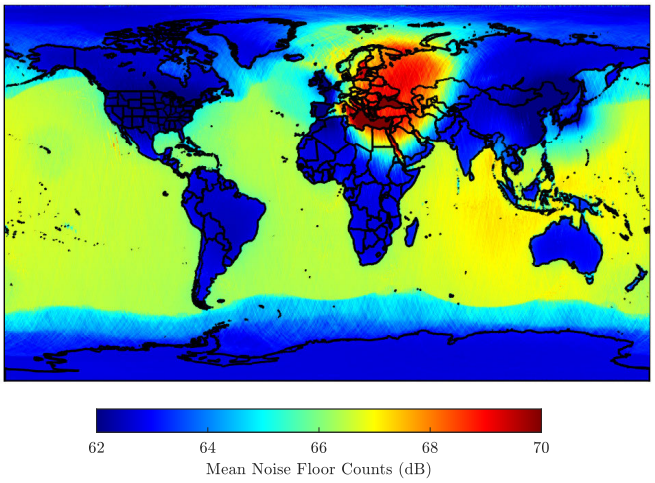
Scott Gleason is with Daaxa LLC, Boulder, CO 80307 USA.

Frederick S. Policelli and Rajat Bindlish are with the NASA Goddard Space Flight Center, Greenbelt, MD 20771 USA.

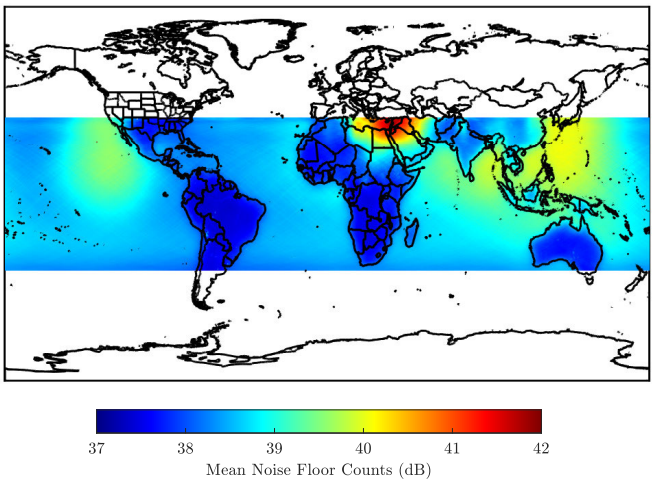
Dorina Twigg and Anthony Russel are with the Space Physics Research Laboratory, University of Michigan, Ann Arbor, MI 48109 USA.

Digital Object Identifier 10.1109/TGRS.2025.3584367

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:56:33 UTC from IEEE Xplore. Restrictions apply.



(a)



(b)

Fig. 1. Mean Level 1 noise floor estimates over the period January 1, 2022–December 31, 2023 from (a) Spire Inc.'s FM146 in polar orbit and (b) CYG01-CYG08 starboard antenna.

subsequent mitigation of their effects are also pertinent, they are not explored in this work.

II. GNSS-R Noise Measurements

A. Operation

GNSS-R systems operate as passive bistatic radar systems, receiving specularly reflected signals off the Earth's surface and later cross correlating them with local "clean" (i.e., distortion free) replicas to form the fundamental GNSS-R measurement, the delay Doppler map (DDM). The DDM partitions power into different delay and Doppler "bins" given the relative motion of the transmitter, receiver, and Earth's surface along with the varied forward propagation paths of received signals. The GPS L1 coarse/acquisition (C/A) signals used in this study are transmitted with a carrier frequency of 1575.42 MHz with a ±1.023-MHz bandwidth. These binary phase shift keying (BPSK) modulated signals use codes of 1023 0.97 μs "chips" that are designed to maximize self-correlation and reduce cross correlations with other GNSS L1 C/A signals. Cross correlation with the appropriate code provides a 30-dB amplification of the desired signal and allows the creation of the DDMs reported in standard Level 1 datasets. Background "noise" levels in Level 1 DDMs can then be determined by computing the power in portions of the DDM known to contain no GNSS signal contributions. Fig. 1(a) and (b) provides the example plots of these noise levels, which are initially recorded in uncalibrated "raw count" units that later undergo several stages of calibration [42], [43], [44]. Background noise measurements are then used to remove their contributions to the total received power [45] in calibrating the reflected signal powers used in the GNSS-R sensing of the Earth.

B. Standard Level 1 Data Trends

Downlinked DDMs (as in the CYGNSS example shown in Fig. 2) are typically a subset of a larger "full" DDM formed onboard the receiver. The onboard tracking algorithm, given a reference surface height and knowledge of the GPS transmitter location, bins received power in the vicinity of a "center bin" (where the peak appears in Fig. 2) corresponding to the delay and Doppler shift of the expected specular point on Earth's surface. Because the specular point is the shortest time delay at which Earth reflections are expected, no Earth reflected power is expected in portions of the DDM at negative relative delays. A portion of the onboard DDM in this negative delay region is used to estimate the combination of internal and external noise contributions associated with the receiver electronics and with any upwelling radiation from the Earth's surface. An average or median power over the selected range is then reported along with the Level 1 data for use in noise compensation.

This noise estimation process is similar to that used in microwave radiometry but occurs in the presence of contributions from multiple GNSS or other navigation satellites operating in the same band that are not selected for a particular observation. The mean noise floor map from the polar-orbiting FM146 Spire Inc. commercial GNSS-R receiver shown in Fig. 1(a) for the period January 1, 2022–December 31, 2023 reveals some of these effects. While ocean surfaces are known to produce lower thermal noise emissions than those of land surfaces (as observed in L-band microwave radiometry), GNSS-observed noise floors are approximately 3.59 dB greater for ocean than land surfaces due to the higher reflections of other GNSS-satellite transmissions associated with water surfaces. Similar trends are also observed using CYGNSS in Fig. 1(b). Other noteworthy trends in the CYGNSS data are "hotspots" in the Pacific Ocean, Timor Sea, Philippine Sea, and East China Sea. The ≈1.5-dB difference relative to the global mean in those cases is associated with transmissions from satellite-based augmentation systems (SBASs) intended to improve GNSS geolocation accuracy. Examples include the Southern Position Augmentation Network (SouthPAN) covering Australia and New Zealand as well as the Wide Area Augmentation System (WAAS) servicing North America and introducing the hotspot in the North Pacific. The trends observed inland, examples of which start in Fig. 1 as far south as Ethiopia and as far north as Finland, are of particular interest in this work. The noise floor increases of up to 8 dB

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:56:33 UTC from IEEE Xplore. Restrictions apply.

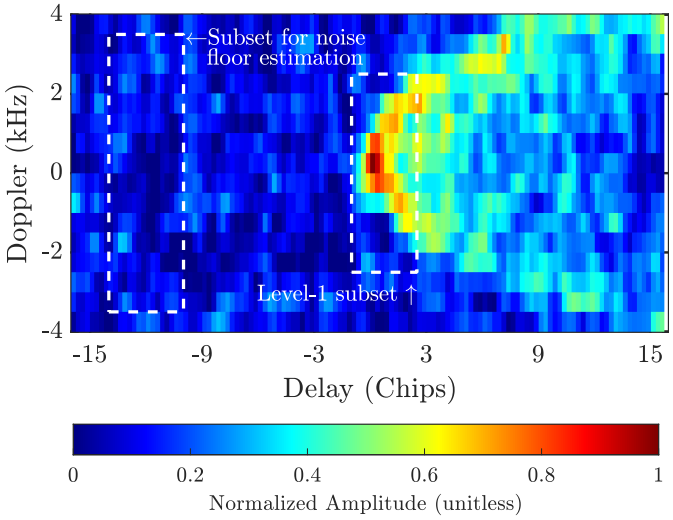


Fig. 2. Measured CYGNSS full DDM, downlinked as part of special acquisition mode, from which standard Level 1 DDMs are cropped and downlinked. The negative delay space corresponds to propagation paths shorter than the specular point and is where an estimate of the measurements' mean noise floors is formed.

above the global land average and the associated spatial trends in such cases appear consistent with the presence of multiple sources of ground-based interference.

C. Raw I/F Acquisitions

While Fig. 1 provides insights into RFI properties, the results shown represent the averages of DDM noise floor quantities that themselves are incoherent averages over 500–1000 ms after correlation with the desired GPS code. Such measurements are not capable of providing information on detailed RFI source properties beyond the average power level received.

Receivers in the CYGNSS constellation can be commanded on occasion to store and downlink "raw intermediate frequency (I/F)" data consisting of the in-phase (I) and quadrature (Q) components of the analog-to-digital converter's measured signal prior to any navigation code correlation. Such data offer increased flexibility in subsequent processing. For example, the data can be cross correlated with a specific transmitted signal with user-selected integration times (e.g., to reduce smearing over land [46]), the phase of measured signals can be extracted and related to properties of the observed surface [47], and signal phases in multiple antennas can be compared [48]. Raw I/F data further allow detailed studies of the signal spectral and statistical properties to detect RFI contributions, in this case prior to any correlation with GPS L1 codes.

When commanded, CYGNSS raw I/F data for a given satellite are recorded for each of the flight module's (FM's) three receive channels (starboard and port for nadir observations and zenith for direct power measurements) simultaneously at ≈16 mega samples per second in each channel. The bandwidths required to store and downlink the data, together with onboard storage limitations, limit each acquisition to no more than a 60-s duration. Since CYGNSS's 2016 launch through the present, ≈500 collections are available that are examined in what follows. Note that the raw I/F mode was initially intended only for calibration and validation diagnostic activities; its utility for other applications has become evident only post-launch.

III. RFI Detection With a Kurtosis Algorithm

A. Method

The kurtosis statistic κ has been shown effective in detecting the presence of RFI in microwave radiometry [21], [22], [25], [26], [27]; its utility for detecting RFI in CYGNSS raw I/F data is examined here. For a random variable s, the kurtosis is defined as

$$\kappa(s) = \frac{\mathbf{E}\left[(s - E(s))^4\right]}{\mathbf{E}\left[(s - \mathbf{E}[s])^2\right]^2} \tag{1}$$

where **E[.]** denotes the expected value operator. For a Gaussian random variable s, $\kappa(s) = 3$; deviations of $\kappa(s)$ from 3, i.e., $\Delta\kappa = |\kappa - 3|$, can then be used as a test for the normality of s. The kurtosis estimator with a finite sample size (in which the E operators are replaced with sample means) for a Gaussian random variable s has a pdf that approaches a Gaussian pdf only as the sample size becomes large. Properties of the estimator are known versus sample size that can be used to set the relationship between a threshold on $\Delta\kappa$ and an RFI detection false alarm rate.

In this work, the kurtosis is evaluated for raw I/F data following a fast Fourier transform over blocks of 2048 samples (a 128 μs time period given the 16-MS/S sample rate). The kurtosis of the real and imaginary values for each of the "bins" in the resulting frequency spectrum is then computed using 1562 spectra (i.e., a period of ≈200 ms). This number of spectra and frequency resolution was selected to promote a more Gaussian kurtosis estimator pdf in the no RFI case while retaining a temporal resolution finer than the mission's Level 1, 2 Hz, data taking rate. The resulting estimates are a frequency-specific, time history of κ or $\Delta\kappa$; these kurtosis spectrograms provide information on RFI properties versus frequency and time.

B. Results

Example results are depicted in Fig. 3 for a December 19, 2022 acquisition by CYGNSS Satellite CYG08. While the acquisition was intended to target a section of the Nile River of interest for hydrological studies, the spectral characteristics observed make the presence of RFI clear. In Fig. 3, the plots of columns 1–3 represent zenith, starboard, and port channel entries, respectively, while rows 1–4 illustrate amplitude spectograms, averages over one second sub-intervals, and the I and Q component $\Delta\kappa$ values, respectively. The 1-s time averages in Fig. 3(d)–(f) are expressed in decibels above a noise floor computed at −3 MHz (i.e., outside the receiver's passband) for each channel. Fig. 3(n) further reports an average power anomaly versus time computed as the difference between the temporally averaged total power in the frequency range $-1 \leq \Delta f \leq 1$ MHz and the "instantaneousness" total powers over the same spectral contents. The frequency Δf is expressed in megahertz relative to the $f_c = 1.575$ GHz center frequency of the GPS L1-band. While the corresponding

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:56:33 UTC from IEEE Xplore. Restrictions apply.

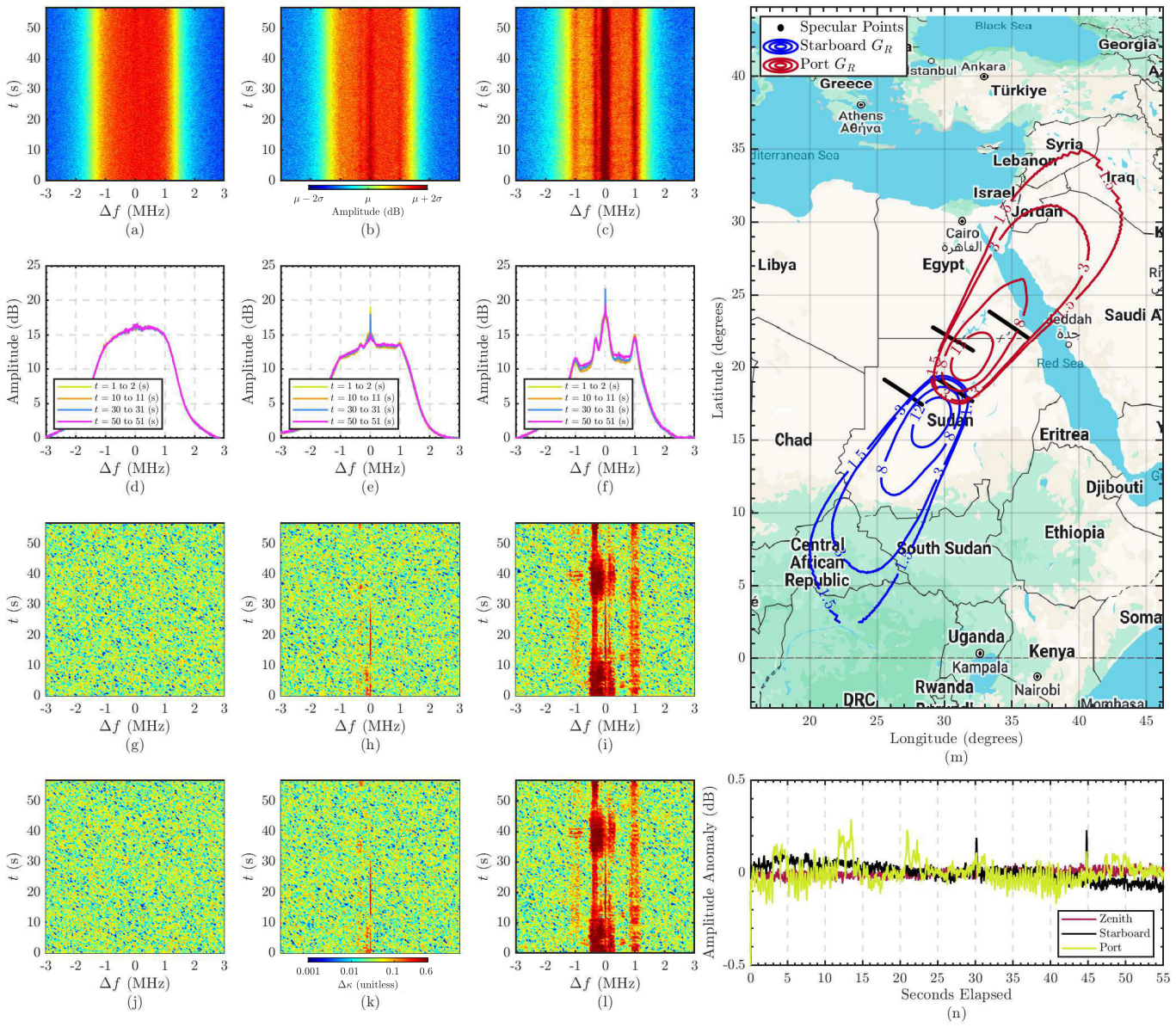


Fig. 3. Kurtosis RFI detector applied to a collection made on December 19, 2022 by CYG08 intended to target a section of the Nile River for hydrological studies. Colorscales of amplitude spectograms are relative to their respective means μ and standard deviations σ. (a) Zenith channel amplitude spectrogram, (b) starboard channel amplitude spectrogram, (c) port channel amplitude spectrogram, (d) zenith channel example spectra at discrete times, (e) starboard channel example spectra at discrete times, (f) port channel example spectra at discrete times zenith channel example spectra at discrete times, (g) zenith channel $\Delta\kappa$ of in-phase component of measured signal, (h) starboard channel $\Delta\kappa$ of in-phase component of measured signal, (i) port channel $\Delta\kappa$ of in-phase component of measured signal, (j) zenith channel $\Delta\kappa$ of quadrature component of measured signal, (k) starboard channel $\Delta\kappa$ of quadrature component of measured signal, (l) port channel $\Delta\kappa$ of quadrature component of measured signal, (m) nadir antenna projections onto the surface at the collection's midpoint time, and (n) time series of amplitude anomalies relative to respective channel temporal median over the $-1.5 \leq \Delta f \leq 1.5$ MHz band across the zenith, starboard, and port channels.

time series in Fig. 3(n) suggests some fluctuations in received power levels throughout the collection's duration, the level of variability is limited to ± 0.25 dB relative to the temporal mean. The map in Fig. 3(m) indicates the locations of the specular points tracked over the acquisition period for GPS reflections; however, the receiver antenna gain contours shown for the port and starboard side antennas are of primary interest for RFI studies.

The L1 GPS signal spectral shape is evident in the zenith channel plot of Fig. 3(d). Relative amplitudes from $\approx$14 to 17 dB occur from -1 to 1 MHz, followed by a gradual tapering of spectral density at the edges of the signal's bandwidth and the CYGNSS receiver's analog passband. In the example shown, both the I and Q $\Delta\kappa$ spectograms for the zenith channel show no apparent evidence of RFI, while clear indications of terrestrial interference occur in both the nadir channel $\Delta\kappa$ spectograms. The port-side response in Fig. 3(f) shows a $\Delta f = 0$ MHz component with a $>$ 20-dB amplitude that is present throughout the acquisition period, along with additional wider bandwidth components spanning the whole of the front end's bandwidth, giving rise to the spectrum's complex shape. In this case, $\Delta\kappa$ is found to be an effective

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:56:33 UTC from IEEE Xplore. Restrictions apply.

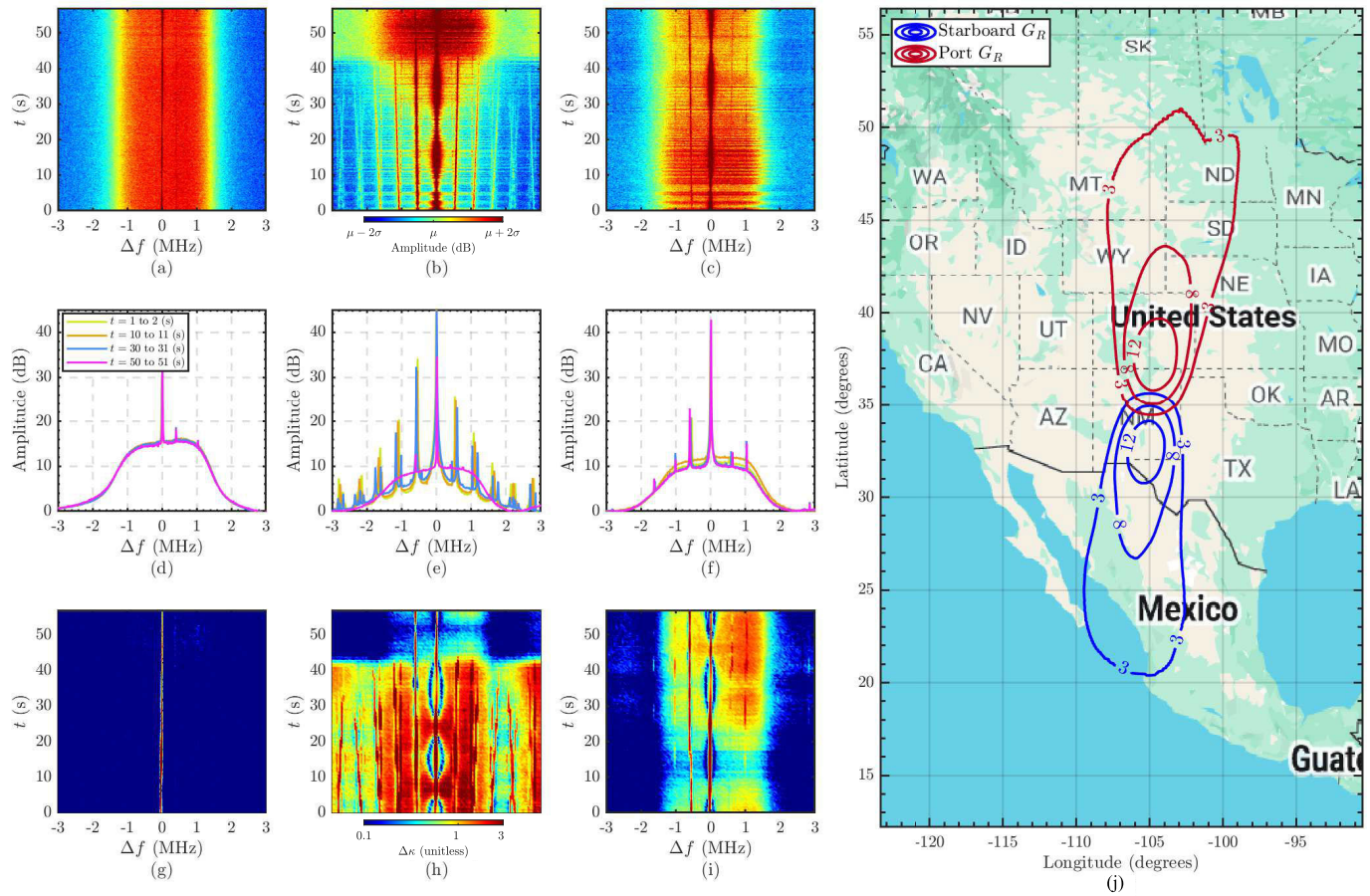


Fig. 4. Kurtosis RFI detector applied to a collection made by CYG05 on August 27, 2021 and was intended to support the CYGNSS mission's soil moisture validation and modeling activities in the San Luis Valley, Colorado, USA. Colorscales of amplitude spectograms are relative to their respective means μ and standard deviations σ. (a) Zenith channel amplitude spectrogram, (b) starboard channel amplitude spectrogram, (c) port channel amplitude spectrogram, (d) zenith channel example spectra at discrete times, (e) starboard channel example spectra at discrete times, (f) port channel example spectra at discrete times zenith channel example spectra at discrete times, (g) zenith channel $\Delta\kappa$, (h) starboard channel $\Delta\kappa$, (i) port channel $\Delta\kappa$, and (j) nadir antenna projections onto the surface at the collection's midpoint time.

detector of an apparent RFI event; Fig. 3(i) and (l) shows the $\Delta\kappa$ values exceeding 0.5, appreciably higher than median zenith channel $\Delta\kappa$ value of 0.034. Additional examinations, including other datasets, suggested that $\Delta\kappa(I)$ or $\Delta\kappa(Q)$ greater than 0.1 for these spectrograms could be taken to indicate RFI, with the RFI in Fig. 3 particularly egregious. The secondary wider bandwidth interference components observed appear to have a "pulsed" nature as evidenced by wider bandwidths of higher kurtosis values observed in Fig. 3(i) and (l) lasting 12 s and then repeating after 18 s. The presence of multiple spectral components associated with this interference source, spectral shape, and periodic nature is all consistent with similar trends reported in [38] using the (Fast, Orbital, Total electron content, Observables, and Navigation (FOTON) receiver deployed aboard the International Space Station (ISS). Murrian et al. [38] report an active interference source in Syria since 2017, which in this case is observed primarily by the CYGNSS port antenna.

Because GNSS-R systems involve bistatic radar observations, a $1/r_T^2 r_R^2$ range loss of signal power typically occurs [49], [50], [51], where r_T and r_R are the range from the specular point on Earth's surface to the transmitter and receiver, respectively. The path loss in contrast for a terrestrial RFI source is $1/r_R^2$ [52], [53] so that >140-dB "amplification" of the RFI signal occurs due to the $r_T = 20\,200$ km range to a GPS transmitter. Murrian et al. [38] further estimate a transmit power level of 79 W for a similar RFI source, so RFI powers that greatly exceed the GPS signal level can occur even when RFI sources are viewed with receive antenna gain levels significantly lower than that of CYGNSS's peak 15-dB gain [note gain contours shown in Fig. 3(m)]. Similar interference effects are observed in the starboard channel at reduced amplitudes, indicating an RFI source whose position lies more within the port antenna footprint. The results also show very similar $\Delta\kappa$ values in both the I and Q components, with similar results for all other tracks, so that subsequent analyses will use the mean of the I and Q $\Delta\kappa$ values.

Fig. 4 illustrates the results from an August 27, 2021 collection conducted by CYG05 that was intended to support the CYGNSS mission's soil moisture validation and modeling activities in the San Luis Valley, Colorado, USA. While the results in Fig. 3 were acquired over a location showing an increased noise floor in Fig. 1, the same plot shows no evidence of increased GNSS-R noise in the Western USA. Values for the San Luis Valley in Fig. 3 are in fact at the global third percentile for Spire Inc.'s FM146 observatory and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:56:33 UTC from IEEE Xplore. Restrictions apply.

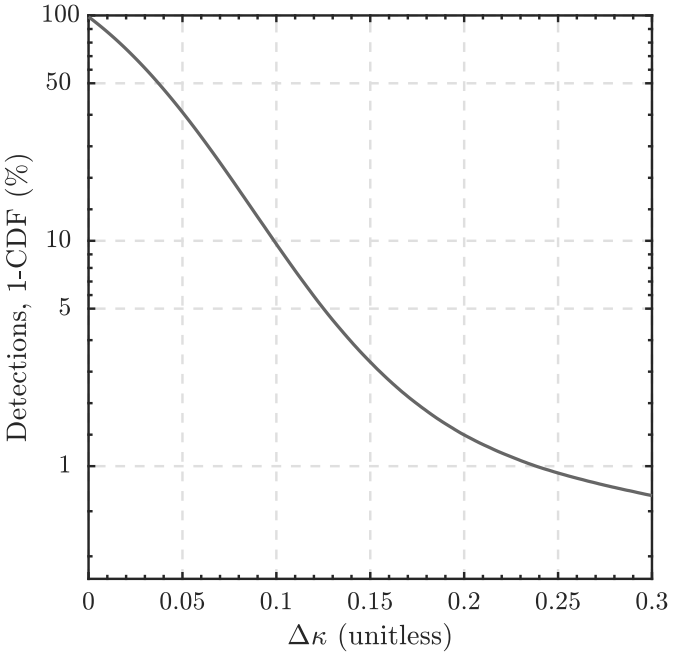


Fig. 5. Percentage of 200-ms intervals in which the maximum $\Delta\kappa$ value within $-1.5 \leq \Delta f \leq 1.5$ MHz exceeded the threshold on the horizontal axis using CYGNSS's complete raw I/F data record.

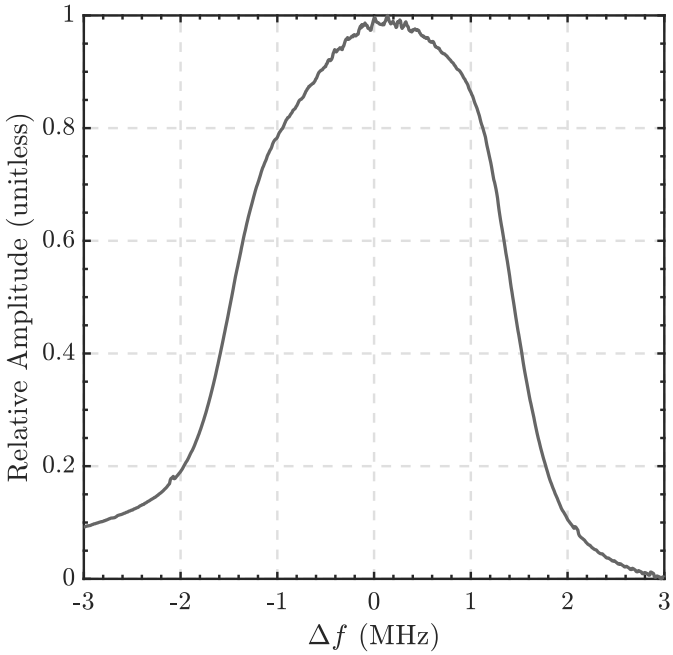


Fig. 7. GPS L1 C/A code signal reference used for cross-frequency RFI detection procedure. The template originates from a zenith channel raw I/F stream collected on January 24, 2018 collection by CYG03.

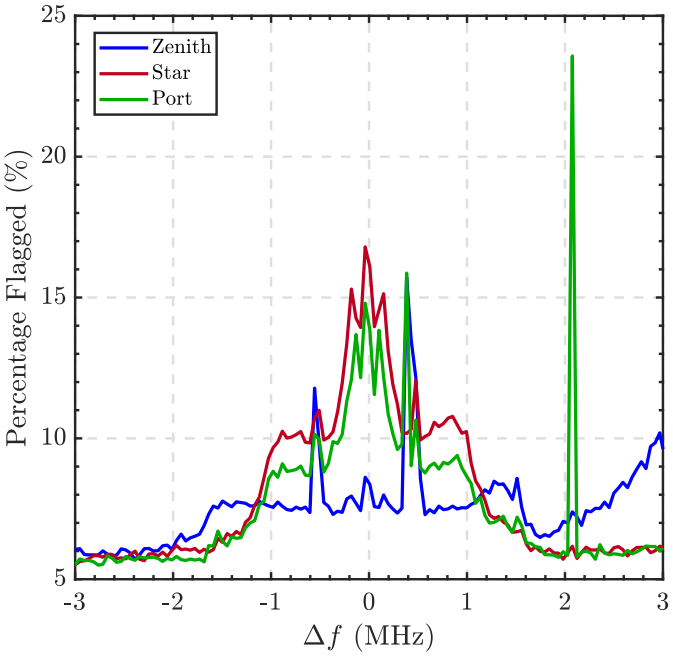


Fig. 6. Percentage of 200-ms intervals detected across CYGNSS's complete raw I/F data record using a $\Delta\kappa = 0.1$ threshold expressed as a function of 50-kHz wide frequency bins.

the fifth percentile for the CYGNSS constellation. Despite this, clear indicators of RFI are observed in Fig. 4. Fig. 4(e), for the starboard channel, shows multiple apparent $\approx$ 200-kHz bandwidth RFI signals whose peak amplitude can be more than 40 dB above the noise floor. Amplitude variations in Fig. 4(b) and (e) similarly reveal a periodic nature in which a narrower band component of the signal lasting 2 s is observed followed by a wider band component lasting 12 s centered about $\Delta f = 0$ Hz, over the first $\approx$40 s of the collection. Multiple persistent tones are also present throughout the acquisition period. The significant change in the signal spectrum in Fig. 4(b) and (e) beginning at approximately 40 s suggests that the starboard channel may have experienced a gain saturation effect that suppressed non-RFI received signal power during the presence of the largest RFI contributions. The resulting kurtosis spectogram in Fig. 4(h) shows $\Delta\kappa$ values exceeding 0.7, clearly indicating RFI across the whole of the receiver's bandwidth that may be impacted by this apparent saturation. Similar but weaker effects are also observed on the port channel. Unlike the example explored in Fig. 3, this interferer also impacts the zenith channel, as evidenced by $\approx$39 dB, $\Delta f = 0$ Hz component despite the antenna's pattern being concentrated away from Earth's surface. Antenna backlobes, multipath effects, or channel cross-coupling within the receiver may contribute to these results.

The preceding examples provide insights into terrestrial RFI source properties and indicate the utility of a kurtosis metric in detecting RFI contributions. While it is also of interest to explore the relative frequency of these occurrences, the capacity for doing so is limited by the absence of a "truth" reference dataset. As an alternative approach, CYGNSS's complete raw I/F data record in all three channels was used to compile the $\Delta\kappa$ statistics summarized in the complementary cumulative distribution function in Fig. 5. The results show that the extreme forms of RFI explored in Figs. 3 and 4 are infrequent. Based on Fig. 5 and examinations of various raw I/F acquisitions, a $\Delta\kappa$ threshold of 0.1 was adopted for identifying RFI, corresponding to an RFI classification for 9.15% of all 200-ms kurtosis spectra examined.

Using this threshold, the frequency of detections versus frequency is plotted in Fig. 6. The results show the zenith

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:56:33 UTC from IEEE Xplore. Restrictions apply.

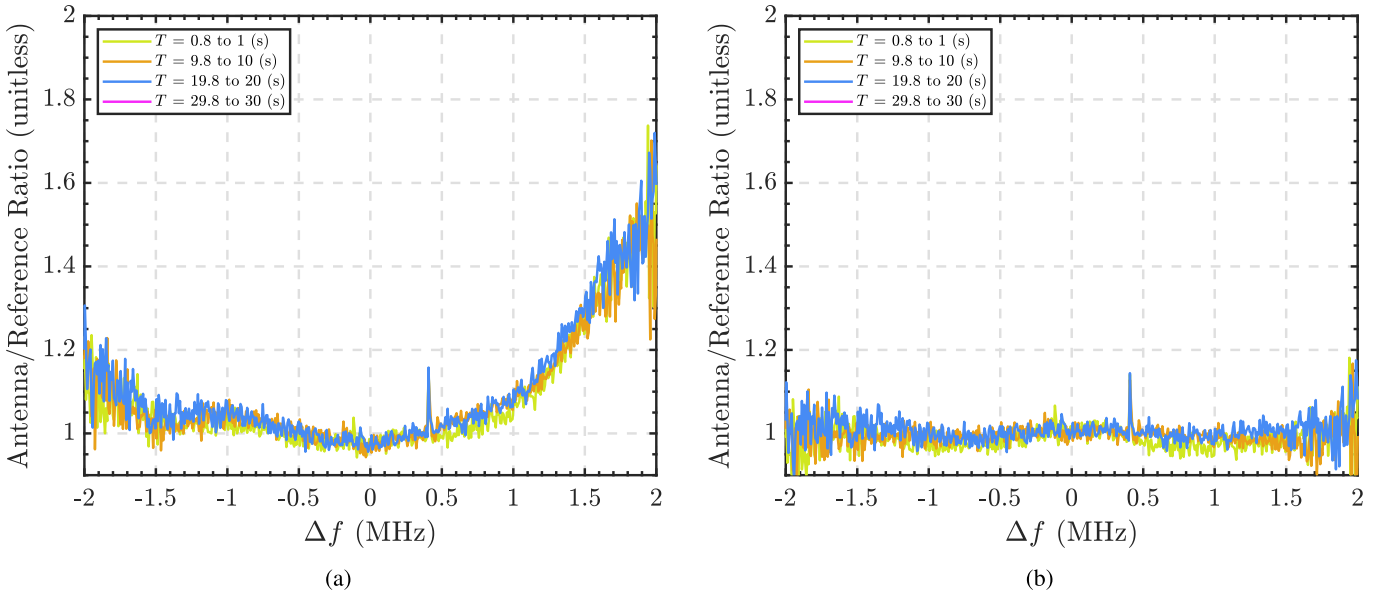


Fig. 8. Example antenna reference "cross-frequency" ratio highlighting practical considerations relating to "residual" non-RFI related ratios associated with receiver front-end bandpass response. (a) Before application of additional flattening parameters. (b) After application of additional flattening parameters.

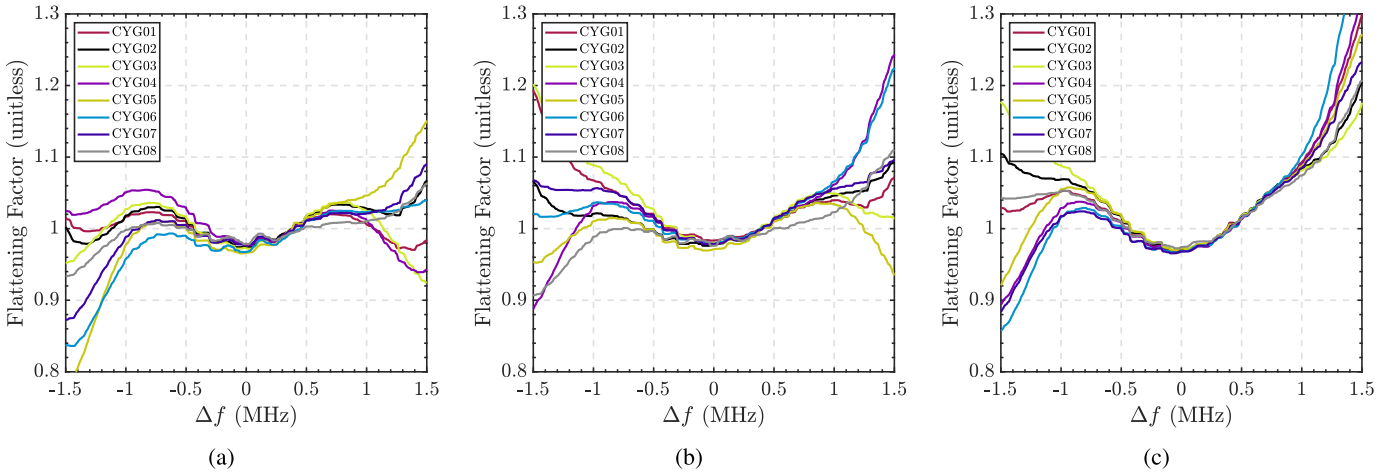


Fig. 9. Additional multiplicative spectral flattening terms derived using in-orbit observations. (a) Zenith. (b) Starboard. (c) Port.

channel to have the lowest percentage of detections (average 7.62%), while the starboard and port channels exhibit similar curves, having average values of 10.14% and 8.93%, respectively. A larger percentage of interferers appear to be transmitting in the $-0.25 \leq \Delta f \leq 0.25$ MHz band with two additional larger values apparent at $\Delta f = 0.38$ MHz and $\Delta f = 2.075$ MHz.

IV. RFI Detection With a Cross-Frequency Algorithm

A. Method

Despite the absence of an independent global reference dataset that is coincident with CYGNSS's ≈500 tracks, the capacity to develop measured signal spectra [see Figs. 3(d)–(f) and 4(d)–(f)] using raw I/F acquisitions allows deviations from the known L1 C/A code spectral shape to provide a robust indication of RFI presence. This, in part, informs the design of the algorithm that follows.

It is commonplace for platforms engaged in RFI detection and mitigation to apply multiple detection algorithms. For example, SMAP's RFI detection approach uses kurtosis, cross-frequency, time-domain outlier, and polarization algorithms at multiple time and frequency resolutions, with a sample flagged as contaminated when detected by any one or more of these detectors.

To explore the use of additional detectors in the context of the RFI events detected by CYGNSS, a "cross-frequency" detection approach [54], [55], [56] was also applied. This approach is premised on the idea that in the absence of RFI, the measured spectrum (S_M) divided by a reference spectrum (S_R) should be nearly constant versus frequency. Defining $\zeta = (S_M/S_R)$, large positive values of $\Delta\zeta = \zeta - 1$ can be taken as indicative of RFI. The reference spectrum shown in Fig. 7 was selected from a January 24, 2018 collection by

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:56:33 UTC from IEEE Xplore. Restrictions apply.

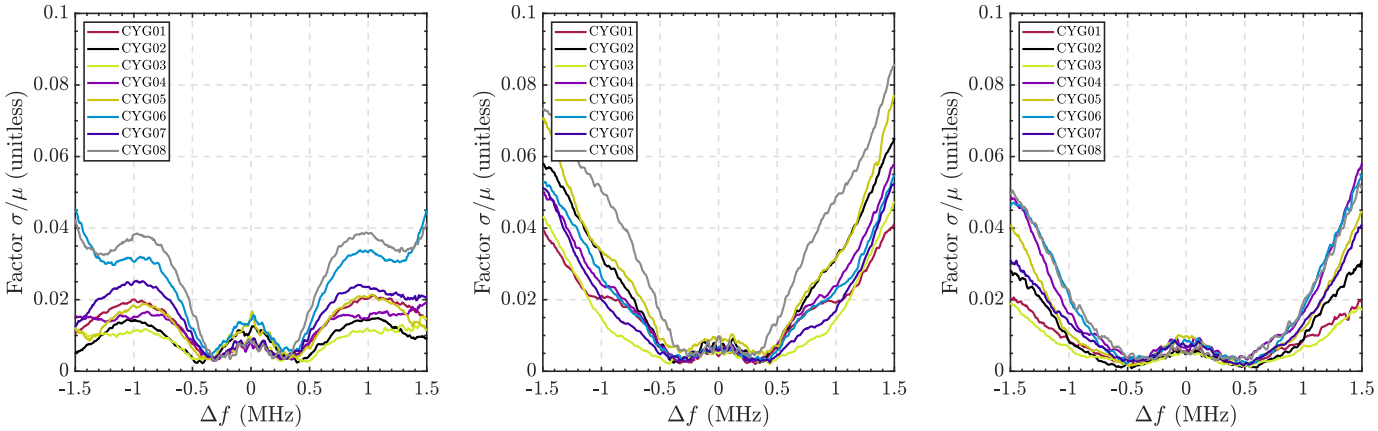


Fig. 10. Uncertainty estimates expressed as solutions' standard deviation relative to the mean highlighting the repeatability of the additional flattening parameters. (a) Zenith. (b) Starboard. (c) Port.

CYG03's zenith channel when the receiver was over South America and believed not to have been impacted by RFI. It is also instructive, for the interpretation of some of the results that follow, to note that the template in Fig. 7 suggests that the 3-dB bandwidth of CYGNSS's front end is on the order of 3 ($\pm$1.5) MHz. Fig. 8(a) plots the example ζ values for a July 7, 2018 acquisition made by CYG01 that are believed to be impacted by an RFI source at $\Delta f = 0.41$ MHz. The curves shown represent 200-ms S_M power integrations originally normalized before dividing by S_R. The interference source observed shows an increase of $\approx$0.15 compared to nearby values. However, other portions of the curve show values up to 1.6 or more that are more likely due to imperfect spectral "flattening" than to RFI. This imperfect flattening result is due to the unique passband shape of each receive channel across the eight CYGNSS satellites. For example, the reference S_R function originates from CYG03, but Fig. 8(a) is applied to CYG01 acquisitions.

Accordingly, an additional flattening procedure was applied analogous to that used for the CubeSat Radiometer RFI Technology (CubeRRT) [56] mission. Because the bandpass response of each of the 24 channels was not characterized prelaunch, a need arises for developing a "best guess" using the subset of measurements believed to be RFI free for each. The procedure for developing these additional flattening parameters is intended to correct for "ratio residuals" and applies to the antenna/reference ratios. For every initial ζ spectogram and within every frequency bin, samples lower than the temporal 20th percentile and greater than their 80th percentile for a given collection are first excluded. Subsequently, a median absolute deviation filter is applied (also known as a Hampel filter [57])

$$\left|s(t) - \Upsilon\left(\tilde{s}\right)\right| \underset{H_0}{\overset{H_1}{\gtrless}} \frac{-1}{\sqrt{2}Q(3/2)} \Upsilon\left(\left|\tilde{s} - \Upsilon\left(\tilde{s}\right)\right|\right) \tag{2}$$

where Υ is the median operator and Q is the inverse complementary error function. This algorithm is applied recursively over windows of 4, 2, 1, and 0.5 s to identify and exclude "global" and "local" anomalies and to exclude outlier contributions in the flattening estimation. Note that $\tilde{s}$ is the vector containing the samples centered about a current sample $s(t)$ and extending over the prescribed duration. For entries that are retained post-filtering, the spectral content is averaged over the duration of the acquisition period. Only spectral components within the first and 99th percentiles are retained, thereby further excluding outliers and developing an estimate of residual ratios and, where necessary, interpolated to fill any spectral gaps. The process is repeated for all acquisitions such that, for each channel and flight unit, $\approx$60 residual reference curves are developed with the final, per channel flattening parameters defined as the inverse of the mean of all available, per channel, residuals. The resulting tables are summarized in Fig. 9 and are complemented with uncertainty estimates defined as the ratio of the standard deviation relative to the mean (σ/μ) of the individual screened, $\approx$60 per channel, curves in Fig. 10. The repeatability of these "in-orbit" derived spectral flattening tables is evidenced by uncertainty levels as low as 0.001 over the frequency range $-1 \leq \Delta f \leq 1$ MHz. Their impact is captured in the example summarized in Fig. 8(b) where they are subsequently applied as a multiplicative factor. Their use achieves more optimal flattening as evidenced by the antenna/references ratios in the frequency ranges $\Delta f \leq -1$ and $\Delta f \geq 1$ MHz. Other practical considerations include the fact that the detection process applies to uncalibrated measurements such that, for every 200-ms "instant," the data are always normalized. The normalization process aims to rescale the RFI free components of any 200-ms "instant" to a 0–1 scale in a similar fashion to Fig 7. To estimate the appropriate normalization parameters for every acquisition and across every individual channel, a procedure similar to that used in the development of Fig. 10 flattening tables is applied in which the first and 99th percentiles are determined for the original frequency spectrum. The first and 99th percentiles are taken to be indicative of the appropriate minimum and maximum to use as part of the normalization process, respectively.

Because the uncertainties depicted in Fig. 10 show that the repeatability of the additional flattening parameters is degraded at increasing separations from the $\Delta f = 0$ Hz center frequency (due to the front end's bandpass filter response roll-off),

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:56:33 UTC from IEEE Xplore. Restrictions apply.

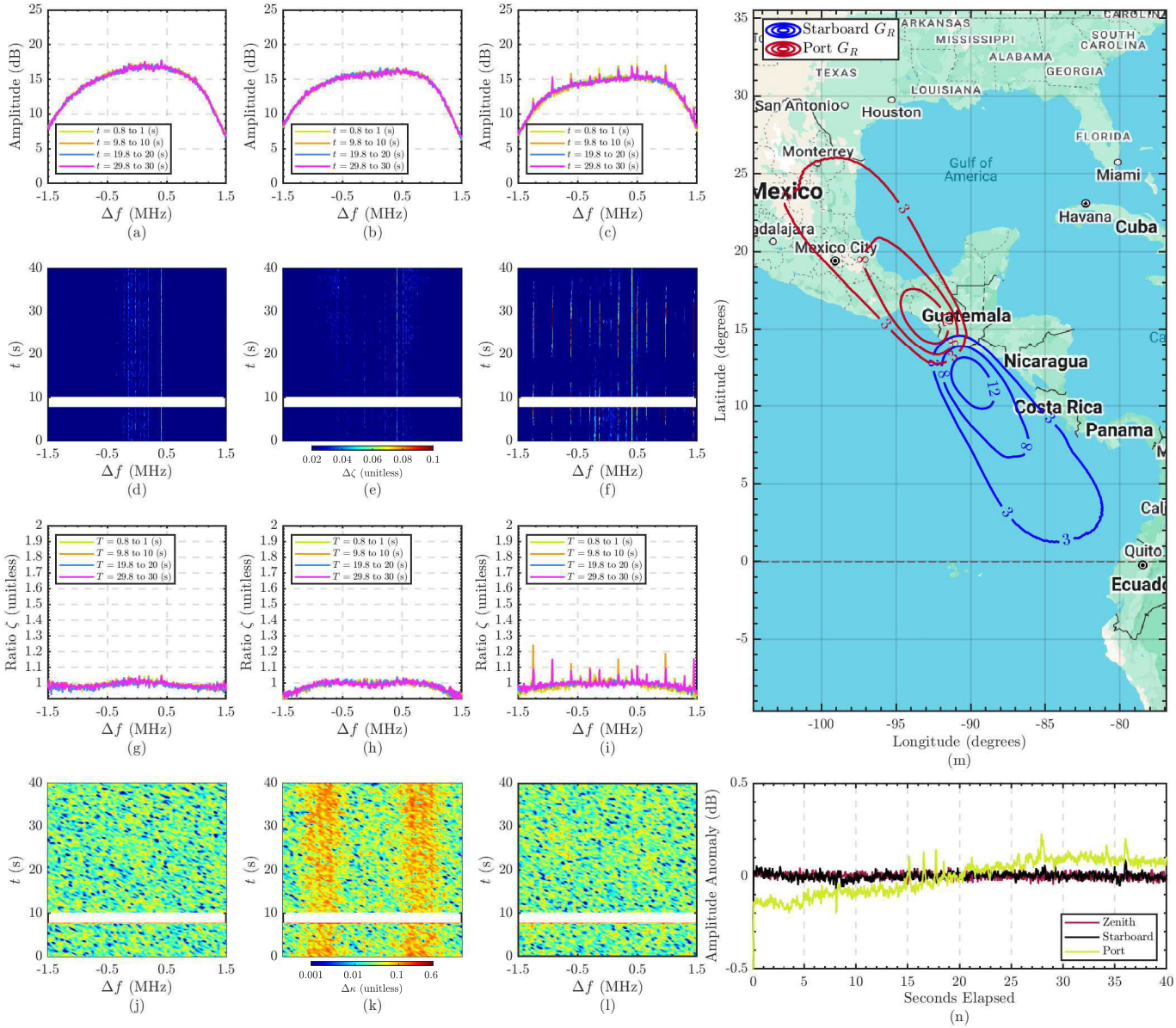


Fig. 11. Cross-frequency detector applied to a collection made on July 13, 2019 by CYG01 intended to target Lake Ilopango, El Salvador, for studies relating to wind vector and wave height retrievals over inland water bodies. Blank sections correspond to blackbody load calibration or DDM receiver resetting intervals. (a) Zenith channel example spectra at discrete times, (b) starboard channel example spectra at discrete times, (c) port channel example spectra at discrete times, (d) zenith cross-frequency $\Delta\zeta$ spectrogram, (e) starboard cross-frequency $\Delta\zeta$ spectrogram, (f) port cross-frequency $\Delta\zeta$ spectrogram, (g) zenith channel example cross-frequency ratio at discrete times, (h) starboard channel example cross-frequency ratio at discrete times, (i) port channel example cross-frequency ratio at discrete times, (j) zenith channel $\Delta\kappa$, (k) starboard channel $\Delta\kappa$, (l) port channel $\Delta\kappa$, (m) nadir antenna projections onto the surface at the collection's midpoint time, and (n) time series of amplitude anomalies relative to respective channel temporal median over the $-1.5 \leq \Delta f \leq 1.5$ MHz band across the zenith, starboard, and port channels.

cross-frequency detections are limited to the frequency range ± 1.5 MHz in what follows.

B. Results

Fig. 11 presents the cross-frequency detector results for a July 13, 2019 collection by CYG01 targeting Lake Ilopango, El Salvador, for studies relating to wind vector and wave height retrievals over inland water bodies [58]. The spectra in Fig. 11(c) make clear that port antenna observations were impacted by RFI, with multiple narrowband components showing $\approx$0.5–1-dB signal amplitudes above the nominal spectrum. The ζ ratios in Fig. 11(f) and (i) show $\Delta\zeta$ exceeding 0.05 in multiple locations. The corresponding port-side kurtosis spectrogram [Fig. 11(l)] shows no $\Delta\kappa$ values greater than the 0.1 standard threshold; a reduced $\approx$0.03 threshold does detect the source but would be associated with a 58.34% flagging rate that clearly would contain a high proportion of false alarms.

Fig. 12 provides a second example from a July 16, 2019 CYG01 collection intended to target Qinghai Lake for studies relating to altimetry and freeze/thaw detection [59]. The cross-frequency detector shows interference in all three receive channels, with the zenith antenna spectogram's peakedness in Fig. 12(c) potentially associated wideband interference

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:56:33 UTC from IEEE Xplore. Restrictions apply.

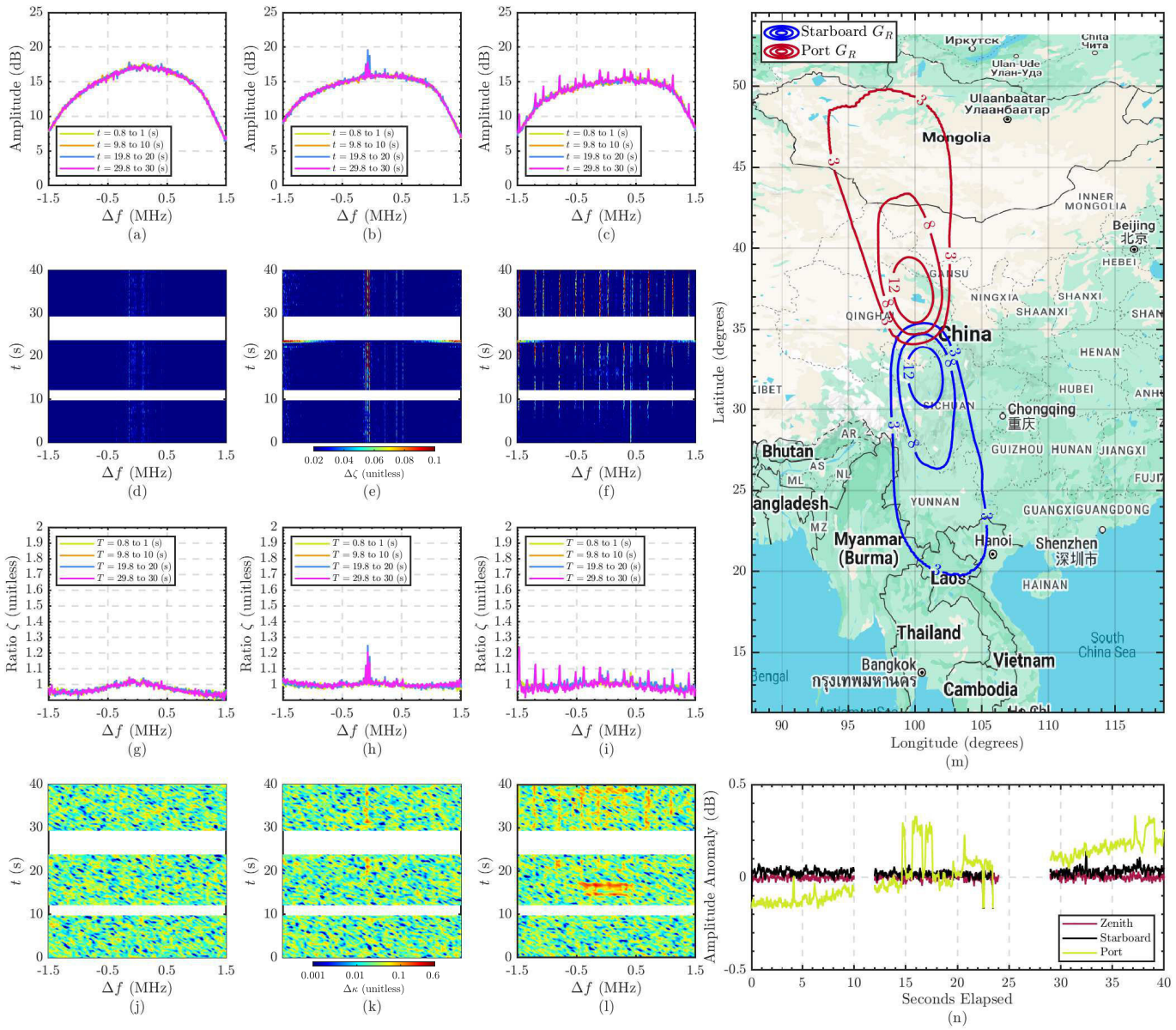


Fig. 12. Cross-frequency detector applied to a collection made on July 16, 2019 by CYG01 intended to target Qinghai Lake for studies relating to altimetry and freeze/thaw detection. Blank sections correspond to blackbody load calibration or DDM receiver resetting intervals. (a) Zenith channel example spectra at discrete times, (b) starboard channel example spectra at discrete times, (c) port channel example spectra at discrete times, (d) zenith cross-frequency $\Delta\zeta$ spectrogram, (e) starboard cross-frequency $\Delta\zeta$ spectrogram, (f) port cross-frequency $\Delta\zeta$ spectrogram, (g) zenith channel example cross-frequency ratio at discrete times, (h) starboard channel example cross-frequency ratio at discrete times, (i) port channel example cross-frequency ratio at discrete times, (j) zenith channel $\Delta\kappa$, (k) starboard channel $\Delta\kappa$, (l) port channel $\Delta\kappa$, (m) nadir antenna projections onto the surface at the collection's midpoint time, and (n) time series of amplitude anomalies relative to respective channel temporal median over the $-1.5 \leq \Delta f \leq 1.5$ MHz band across the zenith, starboard, and port channels.

extending $-0.25 \leq \Delta f \leq 0.25$ MHz. The starboard channel shows a narrowband interference source at $\Delta f = -0.9$ MHz, while numerous narrowband transmissions appear on the port side channel. The $\Delta\kappa$ value for these acquisitions flags RFI only in a few time instants and narrow bandwidths on the starboard and port channels.

These examples were selected to highlight the cross-frequency detector's utility in providing RFI detection capabilities that complement the kurtosis approach, as has previously been noted in microwave radiometry [60], so that both approaches can be valuable when combined. Fig. 13 provides the evidence of this point: $\Delta\kappa$ does not detect the interference observed on the port channel in Fig. 13(i), while $\Delta\zeta$ does not capture the interference impacting the starboard channel.

Through applying the cross-frequency detector to CYGNSS's complete raw I/F data record, the resulting complementary cumulative distribution function capturing the percentage of measurements "detected" as a function of $\Delta\zeta$ estimates is depicted in Fig. 13. The information contained provides information on the RFI environment and may be useful in continued studies for assessing tradeoffs between the threshold enforced and the rate of flagging. For example,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:56:33 UTC from IEEE Xplore. Restrictions apply.

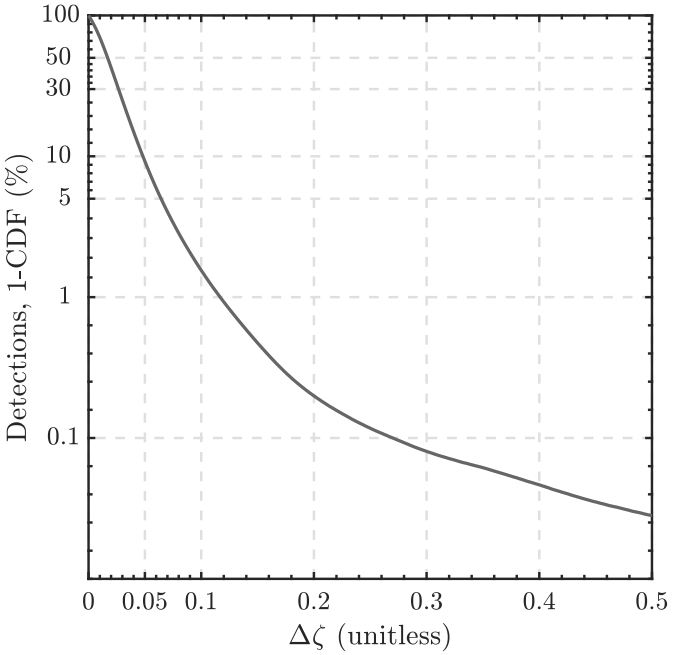


Fig. 13. Percentage of 200-ms intervals detected across CYGNSS's complete raw I/F data record as a function of $\Delta\zeta$ thresholds. The estimates are limited to the $-1.5 \leq \Delta f \leq 1.5$ MHz spectral contents.

TABLE I

SUMMARY OF RAW I/F ACQUISITIONS WHERE $\Delta\kappa$ AND/OR $\Delta\zeta$ SUGGEST THAT MORE THAN 50% OF THE ACQUISITION PERIOD WAS EXPOSED TO TERRESTRIAL RFI SOURCES. *F* IDENTIFIES THE CYGNSS FM AND *C* IDENTIFIES THE NADIR CHANNEL (STARBOARD *S* OR PORT *P*) EXHIBITING THE SPECTRA WITH THE MOST CONTAMINATION (IN TERMS OF RFI PEAK AMPLITUDES ABOVE NOMINAL L1 C/A CODE LEVELS). THE DATE IS IN THE MM/DD/YY FORMAT AND THE LONGITUDES θ AND LATITUDES φ ARE IN DEGREES

F	Date	C	θ	φ	F	Date	C	θ	φ
1	9/9/17	P	-74.9	24.4	6	8/25/17	S	-97.2	23.5
2	8/31/18	S	105.8	9.2	6	9/10/17	S	-79.4	25.4
2	5/19/22	P	-84.1	28.6	6	2/2/18	P	18.4	35.7
3	9/6/17	P	-66.7	18.2	6	3/31/18	S	149.0	26.9
3	9/9/17	P	-77.5	23.2	6	10/12/18	S	-89.9	24.8
3	8/21/19	P	-88.5	13.7	6	9/20/22	S	67.5	17.9
4	10/8/17	S	-89.2	24.9	7	8/25/17	S	-95.0	24.8
4	8/15/19	P	-91.0	11.0	7	10/12/18	S	139.1	12.9
4	8/19/19	P	-108.9	34.0	7	9/20/22	S	101.8	8.9
5	9/9/17	P	-75.3	26.0	8	8/25/17	S	-96.9	23.1
5	1/24/18	S	103.0	2.5	8	2/27/19	S	-87.2	23.8
5	11/9/19	P	-83.9	28.8	8	9/3/22	P	166.5	-15.7
5	2/22/20	P	-82.5	25.3	8	8/25/17	P	123.2	-13.5
5	8/27/21	S	-105.2	32.8	8	10/8/17	S	72.3	17.9

a $\Delta\zeta = 0.025$ threshold identifies 40% of the data as having experienced RFI, while $\Delta\zeta = 0.15$ detects no more than 0.14%. The preceding examples coupled with an analysis of the larger CYGNSS record suggest that thresholds of $\Delta\zeta$ of 0.05–0.075 appear to provide a reasonable balance of detection performance and false alarm rate for the cases considered.

Using both detectors, the results shown were applied on time scales finer than the 0.5 to 1 s standard Level 1 integration durations, thereby providing more insights into shorter time scale behaviors. Across the 498 raw I/F tracks analyzed as part of this study, all are thought to contain at least one 200-ms segment containing RFI, while no more than 25% show contamination that appears significant. To aid future raw I/F users in identifying the acquisitions most affected, those where more than 50% of the collection period across at least 75% of the spectrum is associated with RFI detections are identified in Table I, amounting to a total of 28 collections. Because geolocation of the RFI's source goes beyond the scope of this article, the coordinates of the surface point corresponding to the boresight of the antenna most impacted are provided as a qualitative indicator of location. Future work will consider additional attempts at source geolocation, additional analyses of source properties, and continued analysis of new raw I/F mode collections.

V. CONCLUSION

The results of this work demonstrate the utility of GNSS-R data for the detection of terrestrial interference within the L1-band, particularly using CYGNSS's raw I/F mode acquisitions whose nominal coverage is limited to ±38° latitudes. A kurtosis threshold of $\Delta\kappa = 0.1$ for the processing approach applied identified ≈10% of all time periods as containing RFI. The cross-frequency algorithm after appropriate "flattening" operations was also shown effective at detecting low amplitude (≤2 dB relative to the nominal L1 C/A code signal level) RFI signals and given a $\Delta\kappa = 0.05$ detection threshold flagged ≈16% of all time periods as containing RFI.

Both detectors agree in their "no RFI present" classifications across 75.70% of the time, while both, concurrently, identify RFI over a 3.20% subset of the data. For measurement periods identified as containing an RFI signal by kurtosis, the cross-frequency detector similarly declares RFI being detected over 37.51% of the data. Conversely, for measurement periods identified as containing an RFI signal by $\Delta\zeta$, $\Delta\kappa$ similarly declares RFI being detected over 19.26% of the data. The fact that both detectors identify distinct subsets of interference events highlights the utility of applying them in a synergistic fashion.

It is also instructive to note that Wiseman [62] leverages reports of aircraft's Automatic Dependent Surveillance-Broadcast (ADS-B), containing indicators of the accuracy of their GPS position solutions, to produce a historical record of qualitative "heat" maps that highlight the areas where anomalies are more frequently reported. An increased frequency of anomalies is indicative of the likely deployment of a terrestrial interference source in an aircraft's general vicinity. While the dataset does not constitute a "truth" source as it does not locate interferers and the estimates it provides are contingent on the frequency of air travel in a given region along with the aircraft's flightpath relative to the position(s) of the interference source(s), most RFI events detected in this work occur at locations where work [62] suggests the presence of interference sources. Additionally, examples like Fig. 4 in USA coincide with the issuance of NOTAMs (Notice to Airmen) by the FAA containing information essential to personnel concerned with flight operations that includes potential disruption to navigation signals.

Finally, it is also helpful to note that while the interference signals observed are within the GPS L1-band frequency, they

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:56:33 UTC from IEEE Xplore. Restrictions apply.

may not necessarily corrupt generated DDMs given their orthogonality to the L1 C/A code.

ACKNOWLEDGMENT

This work utilized data made available through the NASA Commercial Satellite Data Acquisition (CSDA) Program.

REFERENCES

[1] M. A. Karegar, T. H. Dixon, R. Malservisi, Q. Yang, S. A. Hossaini, and S. D. Hovorka, "GPS-based monitoring of surface deformation associated with CO_2 injection at an enhanced oil recovery site," *Int. J. Greenhouse Gas Control*, vol. 41, pp. 116–126, Oct. 2015, doi: 10.1016/j.ijggc.2015.07.006.

[2] A. Koulali et al., "GPS-observed elastic deformation due to surface mass balance variability in the southern Antarctic peninsula," *Geophys. Res. Lett.*, vol. 49, no. 4, pp. 1–11, Feb. 2022, doi: 10.1029/2021gl097109.

[3] P. Yuan et al., "An enhanced integrated water vapour dataset from more than 10 000 global ground-based GPS stations in 2020," *Earth Syst. Sci. Data*, vol. 15, no. 2, pp. 723–743, Feb. 2023, doi: 10.5194/essd-15-723-2023.

[4] I. Kudłacik, J. Kapłon, G. Lizurek, M. Crespi, and G. Kurpiński, "High-rate GPS positioning for tracing anthropogenic seismic activity: The 29 January 2019 mining tremor in Legnica-Głogów copper district, Poland," *Measurement*, vol. 168, Jan. 2021, Art. no. 108396, doi: 10.1016/j.measurement.2020.108396.

[5] S. Gleason and V. Zavorotny, "Bistatic radar cross section measurements of ocean scattered GPS signals from low Earth orbit," in *Proc. IEEE Int. Symp. Geosci. Remote Sens.*, Jul. 2006, pp. 1308–1311.

[6] J. Park, J. T. Johnson, A. O'Brien, and S. Lowe, "Studies of TDS-1 GNSS-R ocean altimetry using a 'full DDM' retrieval approach," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, vol. 17, Jul. 2016, pp. 5625–5626.

[7] C. Ruf et al., *CYGNSS Handbook*, 2nd ed., Ann Arbor, MI, USA: Michigan Publishing Services, 2016.

[8] F. Huang et al., "Assessment of FY-3E GNOS-II GNSS-R global wind product," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 7899–7912, 2022. [Online]. Available: https://ieeexplore.ieee.org/document/9882299

[9] M. J. Unwin et al., "An introduction to the HydroGNSS GNSS reflectometry remote sensing mission," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 6987–6999, 2021, doi: 10.1109/JSTARS.2021.3089550. [Online]. Available: https://ieeexplore.ieee.org/document/9456091

[10] A. M. Warnock, C. S. Ruf, A. Russel, M. M. Al-Khaldi, and R. Balasubramaniam, "CYGNSS level 3 merged wind speed data product for storm force and surrounding environmental winds," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 6189–6200, 2024, doi: 10.1109/JSTARS.2024.3379934.

[11] C. Ruf et al., "Characterization of CYGNSS ocean surface wind speed products," *Remote Sens.*, vol. 16, no. 22, p. 4341, Nov. 2024, doi: 10.3390/rs16224341.

[12] L. J. Wernicke, C. C. Chew, and E. E. Small, "Spatially interpolated CYGNSS data improve downscaled 3 km SMAP/CYGNSS soil moisture," *Remote Sens.*, vol. 16, no. 16, p. 2924, Aug. 2024, doi: 10.3390/rs16162924.

[13] O. Eroglu, M. Kurum, D. Boyd, and A. C. Gurbuz, "High spatio-temporal resolution CYGNSS soil moisture estimates using artificial neural networks," *Remote Sens.*, vol. 11, no. 19, p. 2272, Sep. 2019, doi: 10.3390/rs11192272.

[14] M. P. Clarizia, N. Pierdicca, F. Costantini, and N. Floury, "Analysis of CYGNSS data for soil moisture retrieval," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 7, pp. 2227–2235, Jul. 2019, doi: 10.1109/jstars.2019.2895510.

[15] S. H. Yueh, R. Shah, M. J. Chaubell, A. Hayashi, X. Xu, and A. Colliander, "A semiempirical modeling of soil moisture, vegetation, and surface roughness impact on CYGNSS reflectometry data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2020, doi: 10.1109/tgrs.2020.3035989. [Online]. Available: https://ieeexplore.ieee.org/document/9882299

[16] H. Carreno-Luengo, G. Luzi, and M. Crosetto, "Above-ground biomass retrieval over tropical forests: A novel GNSS-R approach with CyGNSS," *Remote Sens.*, vol. 12, no. 9, p. 1368, Apr. 2020, doi: 10.3390/rs12091368.

[17] M. Asgarimehr, D. Entekhabi, and A. Camps, "Diurnal vegetation moisture cycle in the Amazon and response to water stress," *Geophys. Res. Lett.*, vol. 51, no. 19, pp. 1–11, Oct. 2024, doi: 10.1029/2024gl111462.

[18] M. Morris, C. Chew, J. T. Reager, R. Shah, and C. Zuffada, "A novel approach to monitoring wetland dynamics using CYGNSS: Everglades case study," *Remote Sens. Environ.*, vol. 233, Nov. 2019, Art. no. 111417, doi: 10.1016/j.rse.2019.111417.

[19] A. Warnock and C. Ruf, "Response to variations in river flowrate by a spaceborne GNSS-R river width estimator," *Remote Sens.*, vol. 11, no. 20, p. 2450, Oct. 2019, doi: 10.3390/rs11202450. [Online]. Available: https://doi.org/10.3390/rs11202450

[20] FCC, "Table of frequency allocations," Federal Commun. Commission, Office Eng. Technol. Policy Rules Division, Washington, DC, USA, Tech. Rep. 47 C.F.R. § 2.106, 2025.

[21] S. Misra and C. S. Ruf, "Detection of radio-frequency interference for the aquarius radiometer," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 10, pp. 3123–3128, Oct. 2008, doi: 10.1109/TGRS.2008.920371.

[22] D. M. Le Vine, P. de Matthaeis, C. S. Ruf, and D. D. Chen, "Aquarius RFI detection and mitigation algorithm: Assessment and examples," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 8, pp. 4574–4584, Aug. 2014, doi: 10.1109/TGRS.2013.2282595.

[23] M. Aksoy and J. T. Johnson, "A comparative analysis of low-level radio frequency interference in SMOS and aquarius microwave radiometer measurements," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 10, pp. 4983–4992, Oct. 2013, doi: 10.1109/TGRS.2013.2266278.

[24] Y. Soldo et al., "Localization of RFI sources for the SMOS mission: A means for assessing SMOS pointing performances," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 2, pp. 617–627, Feb. 2015, doi: 10.1109/JSTARS.2014.2336988. [Online]. Available: https://doi.org/10.1109/jstars.2014.2336988

[25] Y. Soldo, A. Khazaal, F. Cabot, and Y. H. Kerr, "An RFI index to quantify the contamination of SMOS data by radio-frequency interference," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 4, pp. 1577–1589, Apr. 2016, doi: 10.1109/JSTARS.2015.2425542.

[26] M. Aksoy, J. T. Johnson, S. Misra, A. Colliander, and I. O'Dwyer, "L-band radio-frequency interference observations during the SMAP validation experiment 2012," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1323–1335, Mar. 2016, doi: 10.1109/TGRS.2015.2477686.

[27] A. Bringer et al., "Properties of the RFI environment at 1400–1427 MHz as observed by the soil moisture active/passive mission microwave radiometer," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 7259–7267, 2021, doi: 10.1109/JSTARS.2021.3092996.

[28] Y. Soldo et al., "Location of radio-frequency interference sources using the SMAP L-band radiometer," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 11, pp. 6854–6866, Nov. 2018, doi: 10.1109/TGRS.2018.2844127.

[29] A. M. Alam, M. Kurum, and A. C. Gurbuz, "Radio frequency interference detection for SMAP radiometer using convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 10099–10112, 2022, doi: 10.1109/JSTARS.2022.3223198.

[30] M. S. Braasch, "Another golden anniversary: NAVSTAR, the global positioning system," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 38, no. 1, pp. 4–12, Jan. 2023, doi: 10.1109/MAES.2022.3213231.

[31] T. M. Roberts, T. K. Meehan, J. Y. Tien, and L. E. Young, "Detection and localization of terrestrial L-band RFI with GNSS receivers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5801311, doi: 10.1109/TGRS.2021.3109524.

[32] E. Axell, F. M. Eklöf, P. Johansson, M. Alexandersson, and D. M. Akos, "Jamming detection in GNSS receivers: Performance evaluation of field trials," *Navigation*, vol. 62, no. 1, pp. 73–82, Mar. 2015, doi: 10.1002/navi.74.

[33] F. Huang et al., "Analysis and mitigation of radio frequency interference in spaceborne GNSS ocean reflectometry data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5801712, doi: 10.1109/TGRS.2023.3289998.

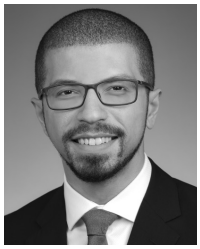
[34] J. Querol, A. Alonso-Arroyo, R. Onrubia, D. Pascual, H. Park, and A. Camps, "SNR degradation in GNSS-R measurements under the effects of radio-frequency interference," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 10, pp. 4865–4878, Oct. 2016, doi: 10.1109/jstars.2016.2597438.

[35] C. Chew, T. M. Roberts, and S. Lowe, "RFI mapped by spaceborne GNSS-R data," *Navig., J. Inst. Navigat.*, vol. 70, no. 4, pp. 1–14, Jan. 2023, Art. no. 618, doi: 10.33012/navi.618.

[36] S. Gleason, J. Johnson, C. Ruf, and C. Bussy-Virat, "Characterizing background signals and noise in spaceborne GNSS reflection ocean observations," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 4, pp. 587–591, Apr. 2020, doi: 10.1109/LGRS.2019.2926695.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:56:33 UTC from IEEE Xplore. Restrictions apply.

[37] D. L. Wu, "GNSS signal jamming as observed from radio occultation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 8642–8645, 2024, doi: 10.1109/JSTARS.2024.3385738.
[38] M. J. Murrian et al., "First results from three years of GNSS interference monitoring from low Earth orbit," *Navigation*, vol. 68, no. 4, pp. 673–685, Dec. 2021, doi: 10.1002/navi.449.
[39] D. Masters et al., "Status and plans for spire's growing commercial constellation of GNSS science cubesats," *Proc. Joint 6th ROM SAF User Workshop 7th (IROWG)*, Konventum, Helsingør, Elsinore, Denmark, Sep. 2019, pp. 1–23.
[40] M. M. Al-Khaldi et al., "An analysis of a commercial GNSS-R soil moisture dataset," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 15480–15493, 2024, doi: 10.1109/JSTARS.2024.3449773.
[41] P. Jales et al., "Spire Global's operational GNSS-reflectometry constellation for Earth surface observations," in *Proc. IGARSS - IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2023, pp. 884–887, doi: 10.1109/igarss52108.2023.10282940.
[42] S. Gleason, C. S. Ruf, M. P. Clarizia, and A. J. O'Brien, "Calibration and unwrapping of the normalized scattering cross section for the cyclone global navigation satellite system," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 5, pp. 2495–2509, May 2016.
[43] S. Gleason, C. S. Ruf, A. J. O'Brien, and D. S. McKague, "The CYGNSS Level 1 calibration algorithm and error analysis based on on-orbit measurements," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 1, pp. 37–49, Jan. 2019.
[44] S. Gleason, M. Al-Khaldi, and D. S. McKague, "Cyclone global navigation satellite system (CYGNSS): Algorithm theoretical basis document level 1B DDM calibration," Ann Arbor, MI, USA, Tech. Rep. 148-0136, vol. 5, Revision 148-0137, 2023.
[45] S. Gleason, M. M. Al-Khaldi, C. S. Ruf, D. S. McKague, T. Wang, and A. Russel, "Characterizing and mitigating digital sampling effects on the CYGNSS level 1 calibration," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5802812, doi: 10.1109/TGRS.2021.3120026.
[46] Y. Wang et al., "Measuring river slope using spaceborne GNSS reflectometry: Methodology and first performance assessment," *Remote Sens. Environ.*, vol. 318, Mar. 2025, Art. no. 114597, doi: 10.1016/j.rse.2025.114597.
[47] W. Li, E. Cardellach, F. Fabra, S. Ribo, and A. Rius, "Assessment of spaceborne GNSS-R ocean altimetry performance using CYGNSS mission raw data," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 238–250, Jan. 2020, doi: 10.1109/tgrs.2019.29361.
[48] M. M. Al-Khaldi, J. T. Johnson, S. K. Chan, and G. Hajj, "Near-specular interferometry with signals of opportunity systems: Potential and limitations," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5102914.
[49] V. U. Zavorotny and A. G. Voronovich, "Scattering of GPS signals from the ocean with wind remote sensing application," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 2, pp. 951–964, Mar. 2000.
[50] A. G. Voronovich and V. U. Zavorotny, "Bistatic radar equation for signals of opportunity revisited," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 1959–1968, Apr. 2018.
[51] A. M. Balakhder, M. M. Al-Khaldi, and J. T. Johnson, "On the coherency of ocean and land surface specular scattering for GNSS-R and signals of opportunity systems," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 10426–10436, Dec. 2019.
[52] T. Wang et al., "Dynamic calibration of GPS effective isotropic radiated power for GNSS-reflectometry Earth remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5800512.
[53] M. M. Al-Khaldi et al., "Quantifying and correcting thermally driven oscillations in CYGNSS real-time GPS effective isotropic radiated power," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5800710.
[54] B. Guner, J. T. Johnson, and N. Niamsuwan, "Time and frequency blanking for radio-frequency interference mitigation in microwave radiometry," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 11, pp. 3672–3679, Nov. 2007, doi: 10.1109/TGRS.2007.903680.
[55] B. Güner, N. Niamsuwan, and J. T. Johnson, "Performance study of a cross-frequency detection algorithm for pulsed sinusoidal RFI in microwave radiometry," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 7, pp. 2899–2908, Jul. 2010, doi: 10.1109/TGRS.2010.2043532.
[56] J. T. Johnson et al., "Real-time detection and filtering of radio frequency interference onboard a spaceborne microwave radiometer: The CubeRRT mission," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 1610–1624, 2020, doi: 10.1109/JSTARS.2020.2978016.
[57] R. K. Pearson, Y. Neuvo, J. Astola, and M. Gabbouj, "Generalized Hampel filters," *EURASIP J. Adv. Signal Process.*, vol. 2016, no. 1, pp. 1–18, Aug. 2016, doi: 10.1186/s13634-016-0383-6.
[58] E. Loria, A. O'Brien, V. Zavorotny, and C. Zuffada, "Towards wind vector and wave height retrievals over inland waters using CYGNSS," *Earth Space Sci.*, vol. 8, no. 7, pp. 1–20, Jul. 2021, doi: 10.1029/2020ea001506.
[59] W. Li, E. Cardellach, F. Fabra, S. Ribó, and A. Rius, "Lake level and surface topography measured with spaceborne GNSS-reflectometry from CYGNSS mission: Example for the lake Qinghai," *Geophys. Res. Lett.*, vol. 45, no. 24, pp. 1–10, Dec. 2018, doi: 10.1029/2018gl080976.
[60] D. M. Le Vine and P. De Matthaeis, "Characteristics of RFI determined from kurtosis using the SMAP radiometer," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5300111, doi: 10.1109/TGRS.2023.3338964.
[61] C. E. Powell, C. S. Ruf, and A. Russel, "An improved blackbody calibration cadence for CYGNSS," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 1002407, doi: 10.1109/TGRS.2022.3165001.
[62] J. Wiseman. (2022). *GPS/GNSS Interference Map*. [Online]. Available: https://gpsjam.org/



Mohammad M. Al-Khaldi (Member, IEEE) received the bachelor's degree in electrical engineering from American University of Sharjah, Sharjah, United Arab Emirates, in 2015, the M.S. degree in electrical engineering from Texas A&M University, College Station, TX, USA, in 2017, the M.S. degree in electrical and computer engineering from The Ohio State University, Columbus, OH, USA, in 2019, and the Ph.D. degree from the Department of Electrical and Computer Engineering and ElectroScience Laboratory, The Ohio State University, in 2020.

He was previously a Project Scientist I, a Post-Doctoral Fellow II, and a Post-Doctoral Fellow I with the University Corporation for Atmospheric Research's (UCAR's) COSMIC Program, Boulder, CO, USA. He is currently a Research Scientist with the ElectroScience Laboratory and the Department of Electrical and Computer Engineering, The Ohio State University. His current research interests include applied electromagnetics, rough surface scattering, and spaceborne remote sensing.

Dr. Al-Khaldi is a member of the NASA CYGNSS Mission Science Team, the NASA CYGNSS Mission Calibration and Validation Team, the IEEE GRSS Technical Committee on Modeling in Remote Sensing, and the IEEE GRSS Technical Committees on Frequency Allocations in Remote Sensing. He is an Associate Member of Commissions B and F of the International Union of Radio Science (URSI) and a NASA CSDA GNSS-R Subject Matter Expert (SME) and serves as an Associate Editor for *Radio Science*.



Joel T. Johnson (Fellow, IEEE) received the bachelor's degree in electrical engineering from Georgia Institute of Technology, Atlanta, GA, USA, in 1991, and the S.M. and Ph.D. degrees from Massachusetts Institute of Technology, Cambridge, MA, USA, in 1993 and 1996, respectively.

He is currently a Professor with the ElectroScience Laboratory, Department of Electrical and Computer Engineering, The Ohio State University, Columbus, OH, USA. His current research interests include the areas of microwave remote sensing, propagation, and electromagnetic wave theory.

Dr. Johnson is a member of Commissions B and F of the International Union of Radio Science (URSI), Tau Beta Pi, Eta Kappa Nu, and Phi Kappa Phi. He received the 1993 Best Paper Award from the IEEE Geoscience and Remote Sensing Society, was named an Office of Naval Research Young Investigator, the National Science Foundation Career Awardee, and the PECASE Award Recipient in 1997, and was recognized by U.S. National Committee of URSI as a Booker Fellow in 2002. He served as the Technical Program Co-Chair for the 2017 International Geoscience and Remote Sensing Symposium. He is also a Past Chair of the GRSS Technical Committee on Frequency Allocations in Remote Sensing. He has served as an Associate Editor for *Transactions on Geoscience and Remote Sensing* since 2000.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:56:33 UTC from IEEE Xplore. Restrictions apply.



Darren S. McKague (Member, IEEE) received the Ph.D. degree in astrophysical, planetary, and atmospheric sciences from the University of Colorado, Boulder, CO, USA, in 2001.

He is currently an Associate Research Scientist with the Department of Climate and Space Sciences and Engineering, University of Michigan, Ann Arbor, MI, USA. Prior to working for Michigan, he worked as a System Engineer at Ball Aerospace and Raytheon and a Research Scientist with Colorado State University, Fort Collins, CO, USA. He is part of the CYGNSS Science Team, working on calibration/validation of the CYGNSS Level 1 and 2 products, as well as part of the CYGNSS Project Management Team. His work has focused on remote sensing with an emphasis on the development of space-borne microwave remote sensing hardware, microwave remote sensor calibration techniques, and mathematical inversion techniques for geophysical retrievals.



Scott Gleason received the B.S. degree in electrical and computer engineering from the State University of New York at Buffalo, Buffalo, NY, USA, in 1991, the M.S. degree in engineering from Stanford University, Stanford, CA, USA, in 1999, and the Ph.D. degree in applied physics from the University of Surrey, Surrey, U.K., in 2007.

He is currently the President of Daaxa LLC, a consulting company, Boulder, CO, USA, specializing in GNSS instruments, calibration, and remote sensing applications. He is a Co-Investigator on the Science Team and an Instrument Scientist for the NASA CYGNSS mission. He has worked in the areas of astronautics, remote sensing, and global navigation satellite systems for more than 20 years, including at NASAs Goddard Space Flight Center, Greenbelt, MD, USA; Stanford's GPS Laboratory, Stanford, CA, USA; Surrey Satellite Technology Ltd., Guildford, U.K.; Concordia University, Montreal, QC, Canada; the Southwest Research Institute; the University Corporation for Atmospheric Research; and the National Oceanography Centre, Southampton, U.K.



Frederick S. Policelli received the Ph.D. degree in Earth and planetary sciences from the Department of Earth and Planetary Sciences, Johns Hopkins University, Baltimore, MD, USA, in 2018.

He studied mechanical engineering and worked in propulsion systems development and testing during his early career. He has extensive experience in applied sciences, systems engineering, and project management. He is currently a Principal Investigator on the NASA mission concept development project and a Co-Investigator on a NASA Applied Sciences funded project that develops near-real-time global-to-regional scale flood mapping products with the goal of extending the results to operational partners for use in disaster management. His research interests include research and applications, including surface water and flood water mapping.

Dr. Policelli is a member of the NASA GSFC Hydrological Sciences Laboratory.



Rajat Bindlish (Senior Member, IEEE) received the B.S. degree in civil engineering from IIT Bombay, Mumbai, India, in 1993, and the M.S. and Ph.D. degrees in civil engineering from The Pennsylvania State University, State College, PA, USA, in 1996 and 2000, respectively.

He is currently with the NASA Goddard Space Flight Center, Greenbelt, MD, USA. Prior to this, he was with the USDA Agricultural Research Service, Hydrology and Remote Sensing Laboratory, Beltsville, MD, USA. He is currently working on soil moisture estimation from microwave sensors and their subsequent application in land surface hydrology. His research interests include the application of microwave remote sensing in hydrology.

Dr. Bindlish is a member of American Geophysical Union. He is a Science Team Member of NISAR, Aquarius, and GCOM-W missions. He is the Deputy Project Scientist for NASA's SMAP mission.



Dorina Twigg received the B.S.E. degree in computer engineering from the University of Michigan, Ann Arbor, MI, USA, in 1991, and the M.S. degree in computer science from Wayne State, Detroit, MI, USA, in 2012.

She is currently a member of the Engineering Staff with the Space Physics Research Laboratory, College of Engineering, University of Michigan, and has a lead role in satellite commanding and scientific data processing with the Science Operations Center for the CYGNSS NASA Earth Venture mission.



Anthony Russel received the B.S. degree in computer science engineering from Michigan State University, East Lansing, MI, USA, in 2014.

He is currently a member of the Engineering Staff with the Space Physics Research Laboratory, College of Engineering, University of Michigan, Ann Arbor, MI, USA. His primary engineering activities involve algorithm development and large-scale data processing as a member of the Science Operations Center for the CYGNSS mission.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:56:33 UTC from IEEE Xplore. Restrictions apply.